

Advanced (Habanero)

Q1. Which number is divisible by 2 and 3?

- (A) 17
- (B) 18
- (C) 19
- (D) 20
- (E) 21

Q2. What is the remainder when 43 is divided by 5?

- (A) 0
- (B) 1
- (C) 2
- (D) 3
- (E) 4

Q3. If $x \cdot 9 = 45$, what is the value of x ?

- (A) 4
- (B) 5
- (C) 6
- (D) 7
- (E) 8

Q4. Which of the following numbers is not divisible by 10?

- (A) 20
- (B) 30
- (C) 40
- (D) 43
- (E) 50

Q5. What is $24 \div 3$?

- (A) 6
- (B) 7
- (C) 8
- (D) 9
- (E) 10

Q6. Which of the following is a multiple of 9?

- (A) 13
- (B) 18
- (C) 22
- (D) 27
- (E) 31

Q7. What is the remainder when 17 is divided by 3?

- (A) 0
- (B) 1
- (C) 2
- (D) 3
- (E) 4

Q8. If $x \div 2 = 7$, what is the value of x ?

- (A) 12
- (B) 13
- (C) 14
- (D) 15
- (E) 16

Open Ended Questions (FRQ)

Q9. Perform the long division $63 \div 5$ and express the result with remainder.

Q10. Prove that a number is divisible by 9 if the sum of its digits is a multiple of 9. Provide an example.

Chef's Tips

- Remind students to apply divisibility rules to simplify calculations
- Encourage students to show their work for long division problems
- Emphasize the importance of expressing division results with remainders